

Hadas Barabash

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EDUCATION

Georgia Institute of Technology	Aug 2020 – Present
• PhD of Computer Science, ACO program	
University of Waterloo	Sep 2025 – June 2025
• Honours Bachelor of Computer Science with Co-op • GPA: 92% Combinatorics and Optimization Minor	

AWARDS

University of Waterloo

Jessie W.H. Zou Memorial Award (2025): \$1,000 for excellence in undergraduate research
National Math Scholarship: (2020-2025) \$25,000 for academic and extra-curricular excellence
Mathematics Undergraduate Research Award (2024): Faculty award for research students
President's International Experience Award (2022): Studied abroad at NTNU in Norway

RESEARCH EXPERIENCE

C&O Department, University of Waterloo. <i>Directed Research Program Mentee</i>	Jan-Apr 2025
- Formalized Tutte's linking theorem, a generalization of Menger's theorem to matroids, using the proof assistant Lean, with two undergraduate students, led by two graduate students - Gave a presentation of our work at an end-of-term Directed Research Program seminar	
C&O Department, University of Waterloo. <i>Undergrad Research Assistant</i>	May-Aug 2024
- Developed and proved approximation algorithms for the NP-complete minimum multicut problem, supervised by Dr. Cheriyan - Used Gurobi and <i>Irs</i> vertex enumeration software to explore lower bounds for the problem's integrality ratio - Presented relevant background for our research at the Canadian Undergraduate Math Conference	

WORK EXPERIENCE

Odaia Intelligence Inc. <i>Data Science Research Co-op</i>	Sep-Dec 2023, Jan-Apr 2022
- Developed classification and regression models leveraging xgboost, scikit-learn, and numPy in Python - Implemented feature engineering for time series prescription data using SQL, python, and pandas - Deployed feature engineering, model training, and inference to AWS Step Functions using terraform - Added Weights & Biases support to streamline model prediction error analysis - Monitored pipeline, triggered containers and tested changes in AWS through ECS and CloudWatch	
MNP <i>Applied Data Intern</i>	May-Aug 2022
- Performed feature engineering, data analysis, and modelling for a project with 40k data points using Python, pandas, xgboost, scikit-learn, and numPy - Used SHAP library for feature explainability and bias detection, improving clients' model understanding - Built front-end for internal web app using React and Javascript, accessing trained model using Flask, and visualising forecasting over time using Chart.js, to communicate model results effectively	
Ford Motor Company <i>Full-stack Software Engineering Co-op</i>	Sep-Dec 2021
- Dynamically created cloud resources in GCP via terraform and google cloud functions in Java - Added functionality using React to internal web application with 100+ users - Gave a presentation on test driven- development with React to 50+ software engineers	

- Using Python with Selenium Webdriver, automated testing of the Oracle Hyperion Planning and Oracle HFM workspace, creating a document recording detailed test results
- Eliminated need for EPM consultants to manually test after software patching

EXTRACURRICULAR

Data Science Club VP of Education and Workshop Lead (2023)

- Managed the education team, including overseeing weekly workshops and research paper reading groups on beginner and intermediate data science topics
- Presented a workshop on the capabilities of AutoML in AWS

National-level Synchronized Swimmer

- Placed 3rd in the duet category in the 2024 Canadian University Artistic Swimming League Nationals
- Placed 1st at Eastern Championships in the 2024 Canadian University Artistic Swimming League
- Placed 4th at the Canadian Artistic Swimming National Qualifier in the Senior Free Duet category
- Competed nationally for 8 years

Math Faculty Orientation Black Tie (2022)

- Helped organize orientation events for incoming students, including ice breakers, games, and scavenger hunts
- Led groups of students through team-building activities

Underwater Hockey Club VP Fundraising (2023-present)

- Organized bake sales to fund club equipment and pool rental
- Organized social events such as a triathlon-style competition with the Quadball and Quiz bowl clubs to promote underwater hockey
- Competed in two North American tournaments

Innertube Water Polo Intramural Team Captain (2022-present)

- Organized and played on intramural innertube water polo team for 7 terms

Underwater Rugby Player (2022)

- Played on NTNU's university underwater rugby team while on a student exchange term in Trondheim, Norway
- Competed in the Norwegian Series